

Automating Standard Laptop Orders with ServiceNow Flow Designer

Team ID: SWTID-2026-4339 | **Date:** 3 October 2026

Note: the uploaded format is written for a MERN stack project. This project is built on the ServiceNow low-code platform, so each MERN heading is mapped to its ServiceNow equivalent (React frontend = Service Portal / catalog; Node.js + Express backend = Flow Designer; MongoDB = ServiceNow tables).

1. Introduction

- **Project Title:** Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
- **Team ID:** SWTID-2026-4339
- **Team Members:** [Name – role, e.g. Flow developer]; [Name – role, e.g. Catalog/UI developer]; [Name – role, e.g. Tester]; [Name – role, e.g. Documentation]

2. Project Overview

Purpose: Automate the ordering of standard laptops so requests move from submission to approval, procurement and asset assignment without manual hand-offs, reducing delays and errors.

Key features:

- Standard Laptop Request catalog item with guided form.
- Rule-based approval (auto-approve within policy, manager/finance otherwise).
- Automatic procurement task and purchase order creation.
- Inventory check before ordering.
- Automated notifications for requester, approvers and fulfilment team.
- Asset record creation and assignment on delivery.
- Reports on turnaround time and pending approvals.

3. Architecture

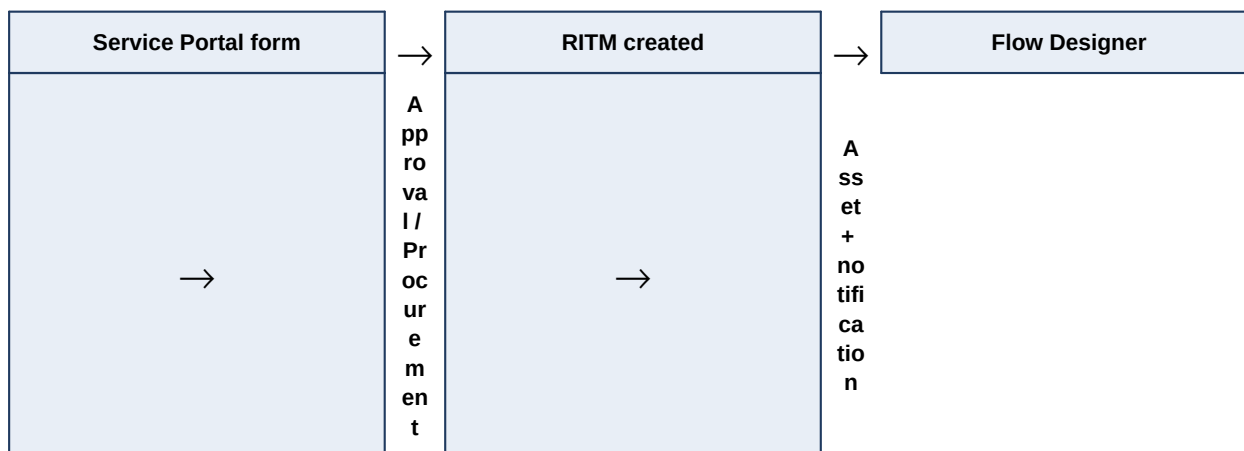
Frontend (Service Portal / Service Catalog): The Standard Laptop Request catalog item with variables (model, justification, delivery location, cost centre), UI policies and client scripts for validation, displayed through the Service Portal.

Backend (Flow Designer): A main flow triggered when a Requested Item is created, calling reusable subflows (Approval, Procurement, Notification) and actions. Conditions, approvals and error handling are configured in the flow, with minimal scripting.

Database (ServiceNow tables): Data is stored in the platform's relational database.

Table	Purpose
sc_request / sc_req_item	Request and requested item created from the catalog
sc_task	Procurement and fulfilment tasks
sysapproval_approver	Approval records for manager/finance
alm_hardware (Asset)	Laptop asset records and assignment

Table	Purpose
proc_po (Purchase Order)	Vendor purchase orders
sys_user / sys_user_group	Requesters, approvers, fulfilment groups



4. Setup Instructions

Prerequisites:

- ServiceNow Personal Developer Instance (developer.servicenow.com).
- Admin access to the instance.
- Plugins: Flow Designer, Service Catalog, Asset Management (as applicable).
- Web browser (Chrome/Edge).

Installation:

1. Request and log in to a Personal Developer Instance.
2. Create an application scope (e.g. Laptop Procurement) in Studio.
3. Create roles and groups: requester, approver, procurement, asset manager.
4. Create the Standard Laptop Request catalog item and its variables.
5. Build the flow and subflows in Flow Designer and activate them.
6. Configure email notifications and test users.
7. Optionally import the provided update set (Retrieved Update Sets > Import XML).

5. Folder Structure

ServiceNow uses application records rather than folders. The application is organized as follows:

Component	Contents
Client (user-facing)	Catalog item, variables, UI policies, client scripts, Service Portal widgets/pages
Server (automation)	Main flow, subflows, custom actions, business rules (if any), script includes
Data	Tables, fields, choice lists, reference qualifiers
Security	Roles, groups, ACLs
Notifications	Email notifications, events, templates
Reporting	Reports, dashboards

6. Running the Application

1. Log in to the instance as a requester test user.
2. Open the Service Portal (/sp) and select Standard Laptop Request.
3. Submit the form; note the request number (REQ/RITM).
4. Open Flow Designer > Operations > Executions to view the flow run.
5. Log in as the approver (if required) and approve/reject.
6. Verify the procurement task, PO and asset record, then confirm request closure.

No separate frontend and backend servers are needed: the platform hosts both.

7. API Documentation

The workflow runs inside the platform, but standard ServiceNow REST APIs can be used to place and track requests.

Method	Endpoint	Description
POST	/api/sn_sc/servicecatalog/items/{sys_id}/order_now	Order the laptop catalog item (body: sysparm_quantity, variables)
GET	/api/now/table/sc_req_item?sysparm_query=number=RITM...	Get requested item status
GET	/api/now/table/sc_task?sysparm_query=request_item=...	List fulfilment tasks
GET	/api/now/table/sysapproval_approver	List approvals
GET	/api/now/table/alm_hardware	List laptop assets

Example request:

```
POST /api/sn_sc/servicecatalog/items/{sys_id}/order_now
{
  "sysparm_quantity": "1",
  "variables": {
    "laptop_model": "Standard Model A",
    "justification": "New joiner",
    "delivery_location": "HQ Floor 2"
  }
}
```

Example response (abridged):

```
{
  "result": {
    "number": "REQ0010001",
    "request_id": "",
    "sys_id": ""
  }
}
```

Replace item names and fields with those used in your instance.

8. Authentication

- **Login:** Users sign in with ServiceNow credentials (or SSO if configured). Sessions are managed by the platform.
- **REST API:** Basic authentication or OAuth 2.0 tokens.

- **Authorization:** Role-based access control with ACLs. Requesters can create and view their own requests; approvers act only on assigned approvals; procurement and asset groups see their own tasks.

9. User Interface

Insert screenshots of: the Service Portal home, catalog item form, request status page, approval screen, and Flow Designer canvas.

10. Testing

Strategy: functional testing of each flow path, role-based testing with test users, and negative testing of invalid input.

- Tools: Flow Designer Test (run flow with test data), Flow Executions log, Automated Test Framework (optional), manual test cases.
- Cases: auto-approval, manager approval, rejection, in-stock, out-of-stock, delivery and asset creation, mandatory-field validation, approval timeout.

11. Screenshots or Demo

Insert screenshots or link: [demo video / link]

12. Known Issues

- Only standard, pre-approved laptop models are supported.
- Vendor ordering is simulated through a PO record unless IntegrationHub is configured.
- Email delivery depends on the instance's mail configuration.
- [Add any bugs found during testing]

13. Future Enhancements

- Extend to monitors, phones and software licences.
- Integrate vendor/ERP systems and budget checks.
- Add Virtual Agent for chat-based requests.
- Add Performance Analytics dashboards and predictive refresh planning.